Global Gateway

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Training Course

Module 1: Introduction to Earth Observation and its Applications for Agriculture

16-20 November 2026

Consortium Lead: GAF

Partners: vito, ramani, Kanalytics Africa Ltd, Caribou

In collaboration with:

Implemented by: esa

Purpose of the week

This opening week introduces Earth Observation (EO) and related geospatial information for agricultural monitoring and climate-smart decision-making in Nigeria. Participants will learn EO fundamentals, explore satellite-derived indicators, interpret agricultural information and assess existing monitoring systems through demonstrations and practical exercises.

Alongside the training, participants will contribute their institutional expertise to the design of the InnoAgri Climate-Smart Information System (CSIS). Starting from real agricultural decisions and existing Nigerian data and capabilities, groups will formulate focused service questions and progressively develop preliminary service canvases. The findings will inform subsequent CSIS design and development, while learning assessments and participant feedback will help tailor the remaining training modules. Workshop ideas are inputs to the project process, not final technical or contractual decisions.

Specific learning objectives

By the end of the course, participants should be able to:

- Explain the basic principles of EO, its main sensor types and their strengths and limitations for agricultural applications.
- Recognise and critically interpret common EO products and indicators, including vegetation indices, time series and anomalies.
- Explore prepared agricultural imagery and geospatial information in QGIS at an introductory level.
- Compare how regional, national and field-level monitoring systems provide useful information to different users.
- Identify relevant Nigerian datasets, existing monitoring tools and institutional capabilities that could support new services.
- Formulate specific agricultural information needs in terms of target users, decisions, spatial scale and timeliness.
- Contribute to a preliminary CSIS service concept, including candidate data, monitoring approaches, functions and delivery options.
- Reflect on learning achievements, future training needs and potential involvement in CSIS development.

Teaching staff

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Timing

The course takes place from Monday 16 to Friday 20 November 2026.

Sessions run from 09:00 to 16:00 on Monday-Thursday. On Friday we may be able to conclude earlier.

Coffee: 10:30-11:00 daily. Lunch: 13:00-13:45.

Daily program

The timetable is indicative. Session content and timings may be adjusted slightly during the week.

Monday 16 November | Welcome and EO foundations

What is InnoAgri, how will we work together, and what can EO contribute to agriculture?

TIME	SESSION	WHAT WE WILL DO
09:00-10:30	Official opening	Welcome remarks by NASRDA, EU Delegation and participating ministries.
10:30-11:00	Coffee break	
11:00-11:40	InnoAgri project and training overview	Project objectives, CSIS vision, training program and Week 1 goals.
11:40-12:10	Participant introduction and survey	Experience, expectations, current knowledge and interests.
12:10-13:00	Introduction to CSIS co-design	How agricultural decisions become focused service questions and canvases.
13:00-13:45	Lunch break	
13:45-14:45	Introduction to Earth Observation	EO principles, sensor types, resolution, strengths and limitations.
14:45-15:30	EO applications in agriculture	Examples in crop monitoring, drought, production and farm decisions.
15:30-16:00	Quiz and discussion	Short Quiz on day 1 contents and Q&A

Tuesday 17 November | Nigerian needs and existing capabilities

Which agricultural decisions need better information, and what can Nigeria build upon?

TIME	SESSION	WHAT WE WILL DO
09:00-10:30	Ministry presentations	Mandates, agricultural decisions, data holdings and information gaps.
10:30-11:00	Coffee break	
11:00-12:15	Workshop: Identify user needs	Individual and group discussion of decisions, target users and unmet needs.
12:15-13:00	Define specific service questions	Cluster broad themes and formulate focused decision-support questions.
13:00-13:45	Lunch break	
13:45-14:30	NASRDA monitoring capabilities	CropWatch and other existing tools, products and operational experience.
14:30-15:30	Workshop: Nigerian data and tools	Inventory relevant EO/non-EO datasets, information services and access constraints.
15:30-16:00	Initial CSIS service canvases	Link questions to users, decisions, existing resources and information gaps.

Wednesday 18 November | EO data, indicators and interpretation

Which EO data and analytical approaches could help answer our agricultural questions?

TIME	SESSION	WHAT WE WILL DO
09:00-09:45	EO data sources for agriculture	Compare satellite missions, available products and their suitability.
09:45-10:30	From observations to indicators	Vegetation indices, time series, anomalies, phenology and interpretation.
10:30-11:00	Coffee break	
11:00-11:30	Introduction to QGIS	Basic GIS concepts, data layers and visualisation.
11:30-12:20	Guided practical: Nigerian EO data	Explore imagery, vegetation indicators and agricultural patterns.
12:20-13:00	Individual practical assignment	Simple map interpretation with optional advanced tasks.
13:00-13:45	Lunch break	
13:45-14:45	Interactive interpretation case studies	Understand crop development, drought and other monitoring signals.
14:45-15:30	Choose suitable data and approaches	Relate EO sources and indicators to the service questions.
15:30-16:00	Service canvas reflection	Record promising approaches, data needs and limitations.

Thursday 19 November | Advanced EO and operational services

How are EO insights combined and delivered through useful agricultural information systems?

TIME	SESSION	WHAT WE WILL DO
09:00-10:15	Recent developments in EO	Cloud processing, data fusion, AI/ML, foundation models, IoT and forecasting.
10:15-10:30	From indicators to services	What an operational monitoring service needs beyond algorithms.
10:30-11:00	Coffee break	
11:00-11:30	Existing monitoring systems: Showcase	Regional early-warning, national monitoring and parcel-level examples.
11:30-12:30	Group evaluation of platforms	Examine users, outputs, capabilities, usability and decision support.
12:30-13:00	Shared lessons from the platform review	Compare useful functions and approaches relevant to Nigeria.
13:00-13:45	Lunch break	
13:45-14:15	InnoAgri proposed capabilities	Review proposed thematic, advisory and delivery components.
14:15-15:30	Workshop: Refine service concepts	Information products, monitoring capabilities, delivery and operations.
15:30-16:00	Shared capabilities across services	Identify common inputs, methods and possible integration.

Friday 20 November | Integration, evaluation and closing

How do the service concepts fit together, and what are the next steps?

TIME	SESSION	WHAT WE WILL DO
09:00-09:40	Final service canvas review	Short group presentations and feedback on proposed services.
09:40-10:30	Towards an integrated CSIS	Explore shared datasets, monitoring components and delivery mechanisms.
10:30-11:00	Coffee break	
11:00-13:00	CSIS prototype investigation	Groups will explore how to get started with the specific CSIS prototypes.
13:00-13:45	Lunch break	
13:45-14:15	Priorities and next steps	Discuss initial development considerations and outstanding questions.
14:15-15:00	Individual learning and feedback survey	Learning achievements, changing CSIS needs and future training interests.
15:00-16:00	Closing and certificates	Summarise results and next steps; certificate ceremony.

How to prepare and what to bring

All participants are requested to:

- Bring a laptop with QGIS installed.
- Register for a free account on the Copernicus Data Space Ecosystem: https://dataspace.copernicus.eu/
- Think of practical agricultural monitoring questions and decisions relevant to your work.
- Bring examples of datasets, reports, monitoring tools or information services used by your organisation, if available and shareable.
- Bring details of colleagues who can answer follow-up questions about data availability and use.

For representatives of the ministries - Tuesday presentation

Please prepare a short presentation (about 20-25 minutes) describing your institution. In particular, address:

- What are your main mandates and which agricultural or livestock decisions do you support?
- What information do you currently use for those decisions, including EO or GIS where relevant?
- Which relevant datasets, surveys, statistics, reports and monitoring systems do you own or use?
- What are the main information gaps or limitations in your present workflows?
- What 2-3 specific questions would you like an improved agricultural information service to answer?

For NASRDA - Tuesday monitoring-capabilities showcase

Please prepare a focused presentation and demonstration (1 hour), including CropWatch and other relevant systems. Address:

- Which monitoring applications and information products are already available, and who uses them?
- Which EO and non-EO datasets, processing methods and infrastructure support these systems?
- How mature is each system (research, prototype, pilot or operational)? How regularly is it updated and who maintains it?
- How are products delivered to users, and are there examples of actual use or validation?
- What are the main gaps, and which data, tools or capabilities could potentially be reused in the CSIS?

Participation, assignments and evaluation

Week 1 is introductory. Activities and assessments are intended to support learning and help tailor subsequent training, not to compare participants with different backgrounds.

WHEN	WHAT WE ASK FROM YOU	HOW IT IS USED
Monday and throughout	Join short polls, knowledge checks and discussions.	Understand expectations and learning progress.
Tuesday	Contribute to needs, capabilities and service-canvas workshops.	Capture institutional knowledge and preliminary CSIS requirements.
Wednesday	Complete a guided QGIS practical and an individual interpretation task.	Assess introductory geospatial and EO interpretation skills.
Thursday-Friday	Review monitoring platforms and develop service concepts in groups.	Document useful monitoring capabilities and service requirements.
Friday	Complete a short knowledge, reflection and course-feedback questionnaire.	Identify learning outcomes, changes in CSIS priorities and future training interests.

Participation, the practical task and the final knowledge/reflection questionnaire will inform the learning assessment. Course completion and certificate eligibility will be communicated by the organisers. Group exercises are collaborative and supported by facilitators.

Course materials and discussion forum

Lecture slides, exercises and reference materials will be available via the InnoAgri training hub. The online discussion forum will support questions to trainers and fellow participants throughout and after the course.